

Aslamzai & Saad (2015) — The Nine Rules

S. Aslamzai and S. Saad, "Pashto Language Stemming Algorithm", *Asia-Pacific Journal of Information Technology and Multimedia* 4(1), 2015, pp. 25-37. Rules and examples transcribed from pages 28-30.

Suffix rules

Rule 1: 5 or more characters, ending in one of these suffixes

→ remove the suffix

Affixes (11): ستان تون ونو تابه ناک وان وال جن ان ور و

Their example: پوهنتون → پوهن پوهن *university*

Their example: موتران → موتر *driver*

Rule 2: 4 or more characters, ending in ي

→ remove ي and add ه

Affixes (1): ي

Their example: پيغله → پيغلي *girls*

Their example: شايسته → شايستي *beautifully*

Rule 3: more than 5 characters, ending in من

→ remove من

Affixes (1): من

Their example: دردمن → درد *sick, patient*

Their example: واکمن → واک *owner*

Rule 4: 5 or more characters, ending in يز

→ remove يز and add ه

Affixes (1): يز

Their example: ټولنيز → ټولنه *humanity*

Their example: دوديز → دود *traditional*

Rule 8: the same word is repeated

→ keep one copy as the root

Their example: خوږي خوږي خوږي → خوږي *sweetly*

Their example: ترخی ترخی ترخی → ترخی *smut*

Prefix rules

Rule 5: more than 3 characters, starting with one of these prefixes

→ remove the prefix

Affixes (10): نا نه م بد نیک تش تل تور تک ډیر

Their example: پوه → ناپوه *uneducated*

Their example: پاتی → تل پاتی *permanent*

Rule 6: 5 or more characters, starting with ال

→ remove ال

Affixes (1): ال

Their example: زیات → الزیات *much more*

Their example: ښه → الښه *very good*

Rule 7: 5 or more characters, starting with وړان

→ remove وړان

Affixes (1): وړان

Their example: خولی → وړان خولی *joker*

Their example: کاره → وړان کاره *bad worker*

Rule 9: more than 3 characters, starting with نا، نه، نیک

→ remove the prefix

Affixes (3): نا نه نیک

Their example: پوه → ناپوه *uneducated*

Their example: روغ → ناروغ *sick*

Their evaluation

Two tests are reported. In the first, the researcher counted correct and incorrect outputs on two samples of 300 words, giving 88% and 89.33%. In the second, 600 stemmed words were shown to two native speakers who judged how many outputs were acceptable, giving 87% and 87.33%. The reported figure of 87.66% is the average.

Points where the paper is inconsistent

Rule 4 says to remove یز and add ه. Its first example does that (ټولنیز → ټولنه) but its second gives دودیز → دود with no ه.

Rule 6 requires five or more characters, but its own example الښه has four.

Rule 3 says more than five characters, but its example واکمن has exactly five.

The Rule 5 prefix list contains a bare م, which would turn موتر into وتر and contradict the Rule 1 example → موتروان → موتر.

The order of prefix and suffix removal is not stated anywhere in the paper.

Rules 5 and 9 overlap: نه، نیک appear in both.